

Omar Essam El-Din Mohamed

Data Engineering & Machine Learning Lead

Technical Skills Portfolio

Programming Languages:

- Python (Advanced)
- SQL (Expert Level)
- JavaScript & HTML/CSS
- PL/SQL

Databases:

- MySQL & PostgreSQL
- Microsoft SQL Server
- Oracle Database
- MongoDB & Firestore (NoSQL)
- BigTable

Data Pipeline & Processing:

- ETL/ELT Pipelines
- Apache Airflow & Kafka
- Apache Spark & MLlib
- Hadoop HDFS
- Data Ingestion & Transformation
- Feature Engineering & Data Mining

Data Warehousing & Storage:

- Data Warehouse Architecture
- Snowflake & Azure Databricks
- Delta Live Tables & Unity Catalog
- Data Modeling & Dimensional Modeling
- EDA & Hypothesis Testing
- Statistical Modeling & Time Series

Amazon Web Services:

- S3 Storage & Lambda
- Bedrock & EC2/ECS

- Azure Databricks
- Data Factory & Functions
- Azure ML Studio

- Compute Engine & GKE
- Cloud Run & Functions
- BigQuery & Vertex AI
- TPUs & Cloud Spanner

Microsoft Azure:

Google Cloud Platform:

Containerization & Orchestration:

- Docker & NVIDIA Docker
- Kubernetes & Microservices
- Container Orchestration

CI/CD & Version Control:

- Jenkins & Git
- Ansible & DevOps Practices
- Unit Testing (pytest, unittest)

Visualization Libraries:

- Matplotlib & Seaborn
- Pandas & NumPy
- Plotly

BI Tools:

- Microsoft Power BI & Tableau
- IBM Cognos Analytics
- Google Data Studio & Excel

Web Frameworks:

- Flask & Django (Python)
- FastAPI & React.js

API Development:

- REST APIs & GraphQL
- API Design & Integration
- Microservices Communication

Generative AI & LLMs:

- Large Language Models (LLMs)
- Retrieval-Augmented Generation (RAG)
- Vector Databases & Embeddings
- Prompt Engineering & Fine-tuning
- PEFT, RLHF, PPO/DPO Optimization
- Instruction Tuning

ML Frameworks & Algorithms:

- TensorFlow & PyTorch
- Scikit-learn & Keras
- Hugging Face Transformers
- LangChain & ONNX
- Supervised/Unsupervised Learning
- Decision Trees, Random Forest, XGBoost

Deep Learning & Computer Vision:

- Neural Networks & CNNs/RNNs
- Transformer Architecture
- Image Classification & Segmentation
- Object Detection & Processing
- Reinforcement Learning

Specialized AI Technologies:

- GPU Computing (CUDA, PyCUDA)
- AI Agent Development
- High-Performance Computing
- Multiprocessing & Distributed Computing
- Model Deployment & MLOps

Enterprise Integration:

- Oracle ERP Systems Integration
- Enterprise Data Architecture
- Business Process Automation
- RPA (UiPath)

Model Deployment & Monitoring:

- Model Serving & Inference
- IBM Watson Studio/ML
- Performance Monitoring & Drift Detection
- Cross Validation & Metrics (RMSE, F1)

